Armadale Kalamunda Group

Orthostatic Hypotension Guideline

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1. Purpose

Orthostatic Hypotension (OH) is common in hospitalised adults (up to 60% of patients) and is positively associated with falls in older adults,

This guideline outlines the assessment, identification, and treatment of OH, including the importance of patient education.

Management of OH is aimed at improving quality of life and reducing symptoms, including the risk of falls, rather than at normalizing blood pressure. Nonpharmacological measures are the key to success.

2. Applicability

Initial identification of OH is achieved through a comprehensive assessment including falls screening, falls history, clinical assessment, medication review and questioning of the patient/family/carer for any description of symptoms.

A lying and standing blood pressure (BP) should be taken by those trained in blood pressure monitoring for the following patients:

- All patients admitted with a fall (where physically able to do so)
- All patients over 65yr of age
- All patients with a delirium
- All patients identified as a falls risk
- All ambulant palliative care patients on admission
- Patients with a medical condition or taking medications that may contribute to OH

3. Guidelines

OH is defined as:

- A drop in systolic BP of 20mmHg of more (with or without symptoms)
- A drop to below 90mmHg on standing even if the drop is less than 20mmHg (with or without symptoms)
- A drop in diastolic BP of 10mmHg with symptoms (although clinically much less significant than a drop in systolic BP).

OH symptoms are caused by a decrease in cerebral blood perfusion and typically occur upon standing during the designated time-period for the drop in blood pressure and resolve upon returning to baseline blood pressure and reclining.

3.1 Symptoms of OH may include

- Falls
- Dizzy/lightheaded, especially when moving from lying or sitting to standing
- Weakness
- Feeling faint or fainting
- Dimming/disturbance of vision
- Sweating and/or nausea
- Gait imbalance
- Mild, momentary confusion
- Unexplained syncope
- Legs buckle

3.2 Causes/contributing factors of OH include

- Frailty and physical deconditioning
- Older adult age
- Standing or sitting up suddenly/quickly
- Hypertension
- Various medical conditions e.g. diabetes, anaemia
- Cardiac conditions including postural tachycardia syndrome, aortic stenosis, arrhythmias
- Medications e.g., diuretics, antidepressants, antihypertensives, antipsychotics, anticholinergics, antiparkinsonian medications, medicines for prostate hypertrophy
- Neurological conditions e.g., Parkinson's disease and some types of dementia
- Endocrinological conditions e.g. Addison's
- Spinal cord injuries
- Autonomic neuropathy
- Dehydration
- Reflex syncope e.g., vasovagal, neurocardiogenic, syncope associated with micturition and defecation
- Exercise
- In the morning when the blood pressure is naturally lower
- Prolonged bed rest or inactivity
- Post prandial

4. Procedure

Taking a lying and standing BP:

- Take twice a day for two days
- Discuss with medical team and discontinue if no evidence/symptoms of OH
- Continue taking lying and standing BP if there is evidence/symptoms of OH

- The BP only needs to be taken at 1 minute standing unless symptoms persist in which case the standing blood pressure must be taken at 3 and 5 minutes.

Procedure:

1. Identify if you are going to need assistance to stand the patient and simultaneously measure the blood pressure (BP).
2. It is preferable to use a manual sphygmomanometer.
3. Explain procedure to the patient.

After at least five minutes	4. The first BP should be taken after the patient has been lying down for at least five minutes.
In the first minute	5. The second BP should be taken within the first minute of the patient standing.

6. Document the result.
7. Symptoms of dizziness, light-headedness, vagueness, pallor, visual disturbance, feelings of weakness and palpitations should be documented on the patient's medical record.
8. Advise the patient of the results and, if the result is positive:
   a. Inform the medical and nursing team.
   b. Take immediate actions to prevent falls and/or unsteadiness.

Results:

A positive result is defined by:

a. A drop in systolic BP of 20mmHg or more (with or without symptoms).
b. A drop to below 90mmHg on standing even if the drop is less than 20mmHg (with or without symptoms).
c. A drop in diastolic BP of 10mmHg with symptoms (although clinically much less significant than a drop in systolic BP).

If results are positive, repeat the procedure regularly or as per medical officer directives.

The patients' falls risk screen, risk identification and interventions required must be reviewed and updated following this procedure.

5. Management of OH

- Educate patients on Isometric exercises prior to going from supine to standing:
  - Lift alternate legs up and down, then move feet up and down
  - Clench and unclench hands
  - Buttock squeezing
- Patients should first sit when going from supine to a standing position

- Provide education (leaflet available) to the patient and or carer and ensure they can manage OH.
- Physical counter-manoeuvres when upright, increases venous return to the heart and enhances orthostatic tolerance:
    - Marching on the spot
    - Leg crossing
    - Standing on tip toes
    - Muscle tensing
- Review of medication contributing to OH and deprescribe offending medicines where possible for Cardiac Heart Failure (CHF) patients, it is preferred to trial reducing doses of ACEI, ARB, ARNI or beta-blockers and diuretics before de-prescribing as these patients require these medications to manage the symptoms of CHF and fluid overload happens quickly and is debilitating.
- Review and update the FRAMP (complete risk assessment and interventions)
- Eating frequent, small meals (post prandial hypotension can occur)
- Full length elastic stocking and consider use of abdominal binders as per medical advice
- Tilting the head of the bed up during the night – approx. 10-20°
- On medical advice may need to reduce salt intake and adhere to a fluid restriction
- Avoiding situations that trigger symptoms, such as standing for long periods
- Lifestyle modification – minimal alcohol
- Consider referral onto appropriate out patient services as indicated on hospital discharge for ongoing management

5.1 Pharmacological treatment

Where possible, medical staff should cease any medications potentially contributing to OH. If pharmacological treatment is required, consider:

- First line: Fludrocortisone (watch for fluid retention, supine hypertension, and electrolyte disturbance)
- In resistant cases, seek specialist advice. Midodrine can be initiated by specialists only.
- Refer to WA state-wide Medicines Formulary for more information. (Timing of dosing is important – the last daily dose should be taken at least 4 hours before bedtime to prevent supine hypertension)

Investigations for medical staff to consider:

When investigating OH, please consider:

- Urea and electrolyte testing (query dehydration)
- Spot morning cortisol to rule out Addison’s disease

6. Legislative context

The following documents are required to give effect to this policy:

- AKG Fall and Fall Injury Prevention and Management Policy
- AKG Comprehensive Care Policy

7. Supporting information

The following documents support the implementation of this policy:

- 'Take 5' in Management of OH
- EMHS OH Patient Information Leaflet
- AKG Fall and Fall Injury Prevention and Management Guideline

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 5 – Action 5.24:
- Standard 6 – Action 6.3:

9. References

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10. Glossary

The following definitions are relevant to this policy.

Orthostatic (Postural) Hypotension	• A drop in systolic BP of 20mmHg of more (with or without symptoms) • A drop to below 90mmHg on standing even if the drop is less than 20mmHg (with or without symptoms) • A drop in diastolic BP of 10mmHg with symptoms (although clinically much less significant than a drop in systolic BP).

11. Document owner

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12. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V1.2	30/04/2024	07/12/2026	Consider referral onto appropriate outpatient services as indicated on hospital discharge for ongoing management

13. Authorisation

This guideline has been authorised and issued by the Director Allied Health

Authorised by	Danielle Kilmurray – Director Allied Health & Ambulatory Care
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